Document No: **Charlie-000-R02**

MAE 434 EXAM 3

Authors: **Samuel Nelson** Date **5/07/2025**
Name

Md Tanzim Ahsan Date **5/07/2025**
Name

Humza Hassan Khan Date **5/07/2025**
Name

Frank Robles Date **5/07/2025**
Name

Approved: **Zackary Malmberg** Date **5/07/2025**
Name of Team Captain

Denotes Change

Revision History

Revision	Revision Date	Description
R00	3/11/2025	Initial Release
R01	4/20/2025	Changes to wing loading and thrust to weight ratio. Changes to initial sizing. Changes to configuration layout and loft. Added special configurations in configuration layout.
R02	5/07/2025	Final revisions for the final project.

Table of Contents

List of Figures

List of Tables

References

1. Developer, O. (2016, January 28). *De Havilland DHC-6 "twin otter"*. Plane & Pilot Magazine. https://www.planeandpilotmag.com/article/de-havilland-dhc-6-twin-otter/
2. Gudmundsson, Snorri (2014). *General Aviation Aircraft Design.* Applied Methods and Procedures
3. *Lax - Los Angeles International Airport*. SkyVector. (2015, February 14). https://skyvector.com/airport/LAX/Los-Angeles-International-Airport
4. Leishman, J. G. (2023, January 1). *Stalling & Spinning*. Introduction to Aerospace Flight Vehicles. https://eaglepubs.erau.edu/introductiontoaerospaceflightvehicles/chapter/maximum-lift-stalling-spinning/
5. Raymer, D. P. (2018). *Aircraft Design: A conceptual approach:* American Institute of Aeronautics and Astronautics Inc.
6. Team, H. I. W. (2017, May 2). *What is the sound barrier?* https://www.howitworksdaily.com/what-is-the-sound-barrier/
7. Husaru, Dorin & Barsanescu, Paul & Zahariea, Danut. (2019). Effect of yaw angle on the global performances of Horizontal Axis Wind Turbine - QBlade simulation. IOP Conference Series: Materials Science and Engineering. 595. 012047. 10.1088/1757-899X/595/1/012047.
8. Williams, R. (2024, December 30). *The versatility of midsize jets: Balancing comfort and cost*. Simple Flying. https://simpleflying.com/midsize-business-jet-guide/
9. Positions brace passengers for impact to reduce injuries ... (1988, February). https://flightsafety.org/ccs/ccs_jan-feb88.pdf
10. USAFacts. (2024, December 16). *Is Flying Safer than driving?* https://usafacts.org/articles/is-flying-safer-than-driving/

Acronyms

NM	Nautical Miles
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulation
ft	Feet
lb.	Pounds
mph	Miles per hour
W_{CREW}	Weight of crew
W_{PL}	Weight of payload
W_F	Weight of fuel
W_E	Empty aircraft weight
C_D	Coefficient of drag
C_L	Coefficient of lift
k_{LD}	Drag efficiency factor
S_{wet}	Wetted surface area
S_{ref}	Reference wing area
CAD	Computer Aided Design
V_{cruise}	Cruise velocity
L	Lift
D	Drag
$C_{L, Takeoff}$	Lift Coefficient (takeoff)

1. Introduction

The purpose of this document is to provide the overall mission profile and flight characteristics of a new business jet that is capable of Short Takeoff and Landing (STOL) and rough terrain landing operations. The document will show multiple conceptual sketches, a selected conceptual design, and initial specifications for sizing, configuration, and payload, as well as wing and tail geometry.

1.1 Project Timeline

To divide the project efficiently between all members, we have organized a Gantt chart representative of the 2 weeks' worth of time we have before our deadline. The Gantt chart is shown below:

PREPARED: <Z. Malmberg M. Ahsan S. Nelson>	DATE: <5/07/2025>	REVISED:	DATE:		Page 9 of 61
CHECKED: <H. Khan>	DATE: <5/07/2025>	CHECKED:	DATE:	TITLE: <MAE434 EXAM 3>	REPORT NO. Charlie-000-R01
APPROVED:< F. Robles>	DATE: <5/07/2025>	APPROVE:	DATE:		REF DWG NO.: N/A

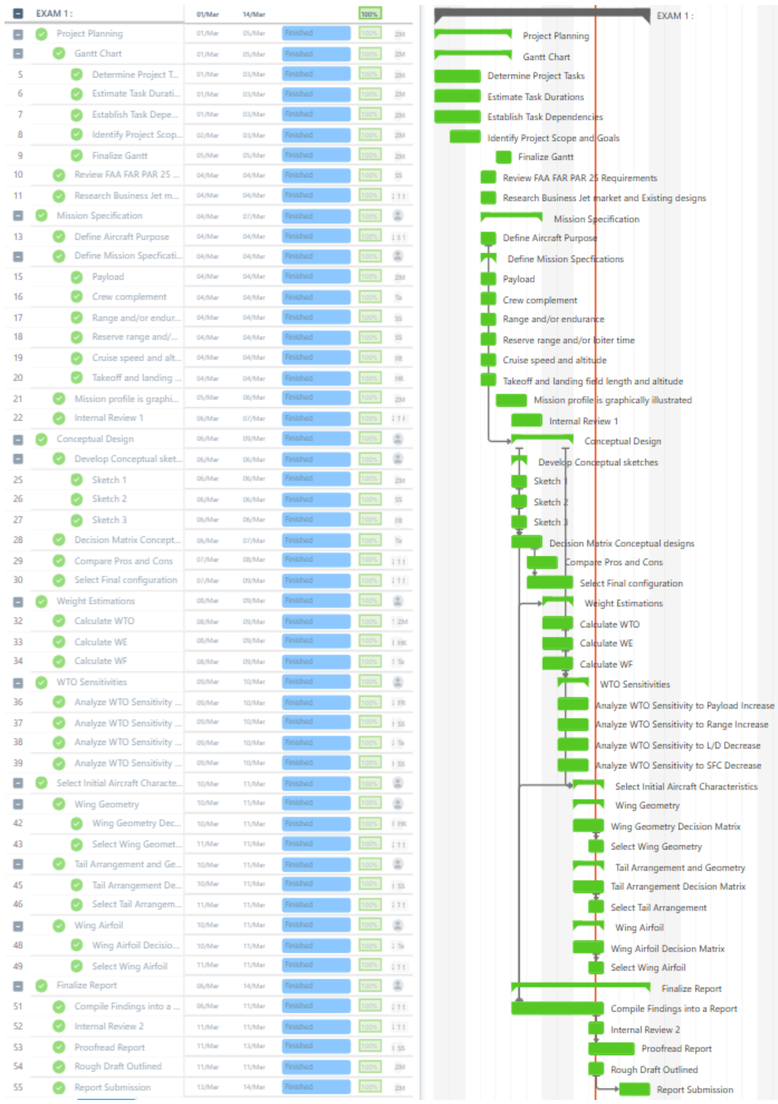

Figure 1: Gantt Chart

2. Mission Specifications

The project requires the team to design a business jet with unique capabilities. For this specific design, the aircraft will have a capacity of 18 passengers, plus a crew complement of two (pilot and co-pilot). The key focus of this aircraft is to access remote and unprepped airstrips while maintaining spaces while offering comfortable accommodation for passengers and cargo. As such, takeoff and landing are limited to a 1200 ft runway. These objectives drive the mission specifications and parameters. The parameters have been listed, and their reasoning has been discussed in the following subsections. Historical data of pre-existing aircraft that serve purposes like the team's design has been considered for deciding on these parameters, such as the de Havilland Canada DHC-6 Twin Otter and the Gulfstream G650. Our aircraft aims to provide the STOL performance of utility aircraft such as the Twin Otter, while offering longer range and speed. Finally, the aircraft will meet FAA FAR 25 requirements for business jets.

2.1 Payload

Based on the standard capacities of current business jets, an 18-passenger space, excluding the cabin crew, was set as a reasonable carrying capacity. The passenger accounts for the average human weight and small luggage weight as total payload weight. As a result, the total payload weight, $W_{PAYLOAD}$, should not exceed 3330 lb. Total payload weight is crucial as it directly derives the range and take-off weight and distance.

2.2 Crew Complement

The aircraft requires 2 pilots for operation and maintenance. This accounts for a crew weight, W_{CREW}, of 370 lb. and is adequate for the proper functioning of the aircraft and while ensuring maximum carrying capacity of passengers. This brings the total capacity of the aircraft to 20.

2.3 Range, Endurance, Cruise Altitude and Speed

The typical range of conventional business jets is about 2000-7000 NM [1]. The design has been made to cover flights from Maine to Los Angeles, roughly 2500 miles. To account for emergencies and access to more airports or landing strips the actual range of the aircraft is 3000 NM. In addition to the range, there will be a reserve for a 45 minute loiter time before proceeding with landing.

At an altitude of 40000 ft, with a speed of sound of 660 mph [2] the aircraft will have a cruise speed of 0.53 Mach or 350 mph. The endurance at this speed is calculated to be 6 hours 18 minutes, with an additional 45 minute loiter time which brings the total endurance to 7 hours 3 minutes.

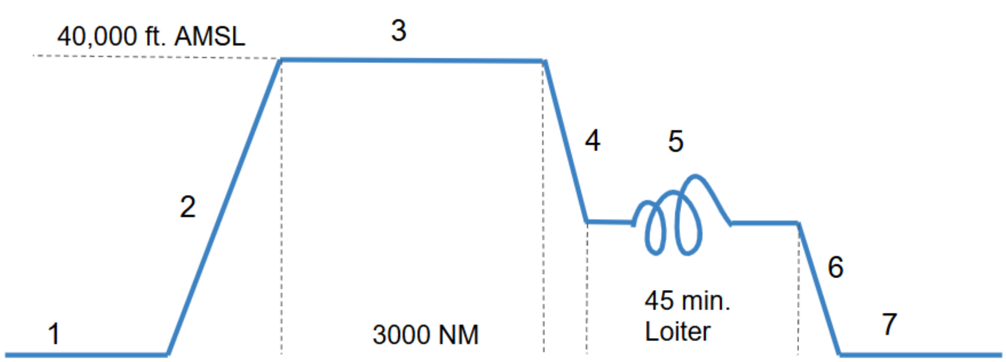

Figure 2: Illustration of Mission Profile

Table 1: Mission Profile

Mission Segment	Segment Name	Segment Description
1	Taxi and Takeoff	Engine startup, warmup, taxi to runway, and takeoff within a 1200 feet runway length.
2	Climb	Climb to a 40,000 feet cruising altitude above sea level
3	Cruise	Cruise at 40,000 feet altitude for 3,000 nautical miles
4,6	Descend	Descend from cruising altitude to airstrip
5	Loiter	Loiter for 45 minutes
7	Landing	Landing, taxis, and engine shut down

Table 2: Altitude Parameters

Parameter	Value	Units
Altitude	40,000	ft
Density	5.87*10^-4	slugs/ft^3
Mach	0.53	
Speed of Sound	968	ft/s
Velocity	513.3	ft/s

2.4 Takeoff & Landing Field Length and Altitude

A key attribute of the design is its capability to take off and land within limited length runways, no more than 1200 ft. This makes the aircraft suitable for landing not only on conventional airport runways but also on small landing strips. This allows more flexibility in the areas it can service, enabling more customer appeal. However, this takeoff and landing field length does vary with altitude of the zone from sea-level to 5000 feet. The LAX, PWM, and BGR airports are within this range of altitude [3].

3. Conceptual Sketches

3.1 Concept 1 (T-tail, Over-Wing Engine)

Concept 1 features conventional fuselage, high mounted wing, and over-wing mounted engines. Over-wing mounted engines are used for blown flaps, which can reduce takeoff and landing distances by increasing C_L over a portion of the wing. A high wing configuration lends itself well to off-airport operations, by providing greater clearance between the wing and the ground and thereby reducing the chance of debris being ingested into the engines. This engine placement suggests the use of a T-tail, to move the horizontal stabilizer out of the jet wash.

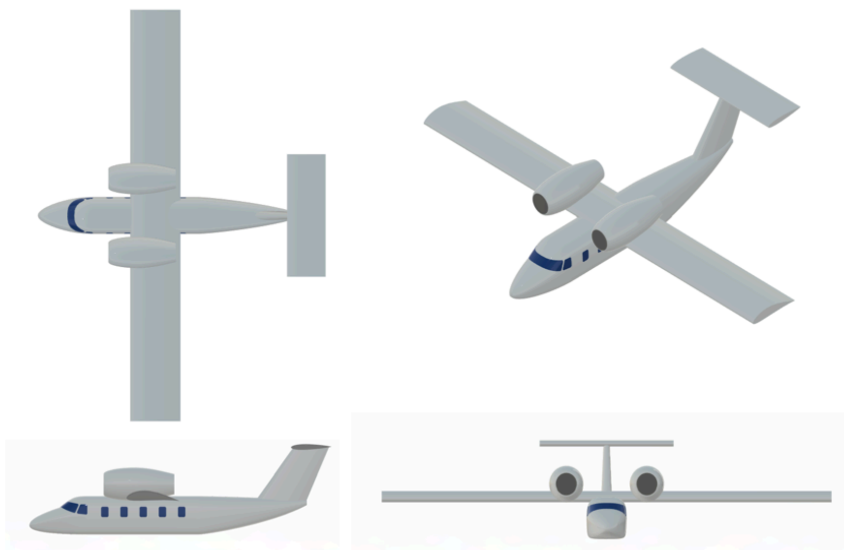

Figure 3: Concept 1 (T-tail, Over-Wing Engines)

3.2 Concept 2 (H-tail, Rear Engines)

Concept 2 features conventional fuselage, low wing, and a H-tail design, and rear over-fuselage mounted engines, like an A-10 Thunderbolt II. The low wing placement shields the engines from debris ingestion upon taking off or landing from a loose surface, but it reduces the ground clearance of the aircraft. Engine placement suggested the use of H-tail configuration to keep the horizontal stabilizer out of the jet wash, while also adding redundancy by having two vertical stabilizers. H-tail can also have a lower cross-sectional area and overall smaller construction, which can lead to reduced weight.

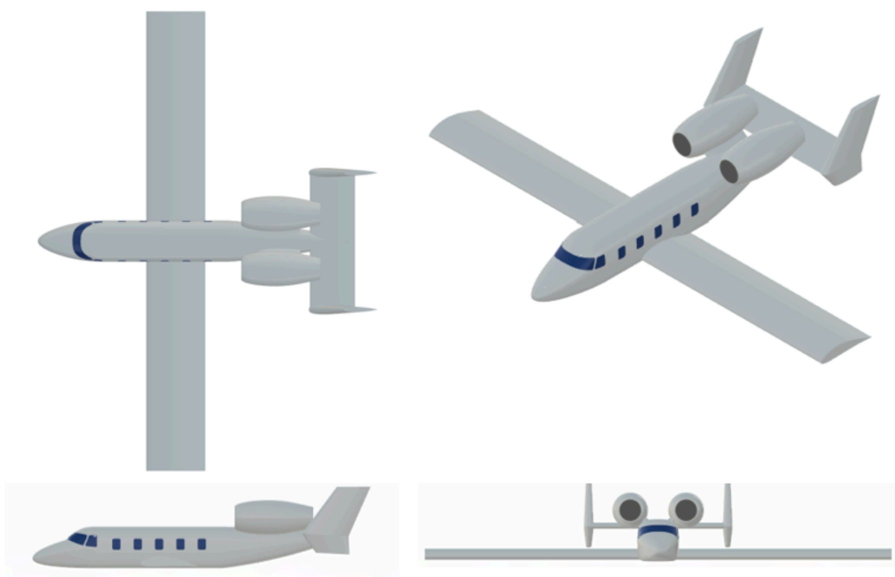

Figure 4: Concept 2 (H-tail)

3.3 Concept 3 (Canard)

Concept 3 consists of a conventional fuselage, a canard configuration, over-wing mounted engines, and twin wingtip mounted stabilizers. Canards allow better stall characteristics, since the main wing never fully stalls. Wingtip-mounted vertical stabilizers double as wing fences, reducing induced drag. However, drawbacks such as less flap control and center of gravity adjustments compared to other concepts make this design less suitable.

Figure 5: Concept 3 (Canard)

Table 3: Conceptual Sketches Decision Matrix

Category	Concept 1	Concept 2	Concept 3
Concept Name	T-tail	H-tail	Canard
Takeoff Performance(5)	5	3	3
Drag(3)	3	3	3
Public Perception(2)	4	3	3
Debris Ingestion(4)	5	4	4
Stall Characteristics(4)	2	3	4
Engine Noise(2)	2	3	3
Total	74	64	68
Rank	1	3	2

3.4 Selected Design

Table 3 compares the three conceptual sketches where each of the six criteria has different weightings that is multiplied to the score given to the concept and adds to its total score. The optimal choice was found to be the T-tail, Over-Wing Engine configuration from concept one. According to our mission requirements, the T-tail, Over-Wing Engine aircraft should achieve the parameters set to ensure the jet aircraft can successfully conduct short takeoff and land in remote areas. However, it noted that none of these designs go into manufacturability difficulties. We are assuming each aircraft would have similar manufacturability challenges, so it was unnecessary to evaluate this in the decision matrix. Also, the drag criteria are about the same for all of them. Each concept has an argument for why it should have a higher drag than the other, but this would be impossible to confirm unless further analysis is done so the group assumed all conceptual sketches produce similar drag values.

4. Initial Weight Estimations

With the finalized configuration, the aircraft's total takeoff weight can be estimated in a MATLAB code with several constants guiding the estimation process for total takeoff weight. The process began with Equation 1 which shows how the total takeoff weight can be rearranged into an implicit formula. Weight of crew (W_{CREW}), weight of payload (W_{PL}), and fuel fraction ($\frac{W_F}{W_O}$) are derived from the mission specification.

$$W_O = \frac{W_{Crew} + W_{PL}}{1 - \left(\frac{W_F}{W_o}\right) - \left(\frac{W_E}{W_o}\right)} \qquad \text{Eq.1}$$

W_{CREW} and W_{PL} were already defined by the mission specifications (370 lbs. and 3330 lbs.). Fuel Fraction can be derived from the mission profile on Figure 2. From Eq. 3, fuel fraction product ($\frac{W_x}{W_O}$) can be further derived from the individual fuel fractions corresponding to each of the different mission phases (Takeoff, Climbing, Cruising, Loiter, Landing). Eq. 2 takes the mission fuel fraction and adds a 6% allowance for reserve and trapped fuel to get a total fuel fraction for the original implicit formula.

$$\frac{W_F}{W_0} = 1.06\left(1 - \frac{W_X}{W_0}\right) \qquad \text{Eq.2}$$

$$\frac{W_x}{W_0} = \frac{W_1}{W_0}\frac{W_2}{W_1}\frac{W_3}{W_2}\frac{W_4}{W_3}\frac{W_5}{W_4} \qquad \text{Eq.3}$$

Going over the individual fuel fraction of each mission phase: Takeoff $\left(\frac{W_1}{W_0}\right)$, climb$\left(\frac{W_2}{W_1}\right)$, and land $\left(\frac{W_5}{W_4}\right)$ fuel fractions can be assumed as 0.97, 0.985, 0.995, respectively. Cruising and loiter fuel have their own function as shown in Eq. 4 and 5. Once the aircraft reaches the cruise stage, the fuel fraction is evaluated using the Breguet endurance equation. Range of flight (R) remains as 3000 NM while Specific Fuel Consumption of the aircraft can be pulled from Raymer's Table 3.3 where the high bypass turbofan propulsion system constrains the value to be 0.4 (1/hr.) during loiter and 0.5 (1/hr.) during cruise. Endurance in Eq.5 will correlate to the total loitering time spent.

$$\frac{W_3}{W_2}_{cruise} = exp\left(- \frac{RC}{V\left(\frac{L}{D}\right)_{cruise}}\right) \qquad \text{Eq.4}$$

$$\frac{W_4}{W_3}_{Loiter} = \left(- \frac{EC}{\left(\frac{L}{D}\right)_{MAX}}\right) \qquad \text{Eq.5}$$

Finally, the max lift to drag ratio can be substituted to Eq.6 where the relationship of aspect ratio, surface wet and reference area, and the K_{LD} constant. Using Raymer's Table 3.6, K_{LD} is 15.5 as our configuration design resembles a civil jet. In terms of Aspect Ratio, 8 was chosen as a guess based on historical data. The surface wet and reference area of the aircraft was taken from the 3D CAD sketch of the model configuration chosen which resulted in a value of 4.3. Using all these values, $\left(\frac{L}{D}\right)_{MAX}$ was calculated to be 21.0442 and $\left(\frac{L}{D}\right)_{Cruise}$ = 18.22 (derived from Eq.7)

$$\left(\frac{L}{D}\right)_{MAX} = K_{LD}\sqrt{\frac{A}{S_{wet}/S_{ref}}} \quad \text{Eq.6}$$

$$\left(\frac{L}{D}\right)_{Cruise} = 0.866\left(\frac{L}{D}\right)_{MAX} \quad \text{Eq.7}$$

The last term in the initial implicit formula, $\left(\frac{W_E}{W_o}\right)$ can be approximated with Eq. 8 where coefficients A and c were identified from Table 3.1 (Raymer) using the jet transport model and K_{VS} is 1 due to the constant wing sweep.

$$\frac{W_E}{W_0} = AW_0^{\ C}K_{VS} \quad \text{Eq.8}$$

4.1 Takeoff Weight Sensitivities

The initial implicit formula (Eq. G.1) has been converted to a function with constants with only W_0 left on either side of the equation. Using the De Havilland Twin Otter Classic 300-Ginitial total takeoff weight value as the first initial guess, the MATLAB code ran a while case iteration to find a suitable takeoff weight where the guess is imputed to the right-hand side of the equation to solve for a "calculated" takeoff weight. A tolerance value of 0.01% was established to ensure that the value of the estimated takeoff weight is close to the value of the calculated takeoff weight. The results of the iterations can be seen in Table 4 below:

Table 4: Weight Results

	Weight Results (lbs)
Total Takeoff Weight	25701.7
Empty Weight	7747.92
Fuel Weight	17956.3

5. Initial Selections

5.1 Wing Airfoil Selection

The wing airfoil was selected based primarily on stall characteristics and max C_L. The NACA 4415 airfoil was chosen, as it provides a gentile stall and high lift coefficient, which are favorable for STOL operation. In order to meet STOL requirements, high lift devices will be implemented, discussed below.

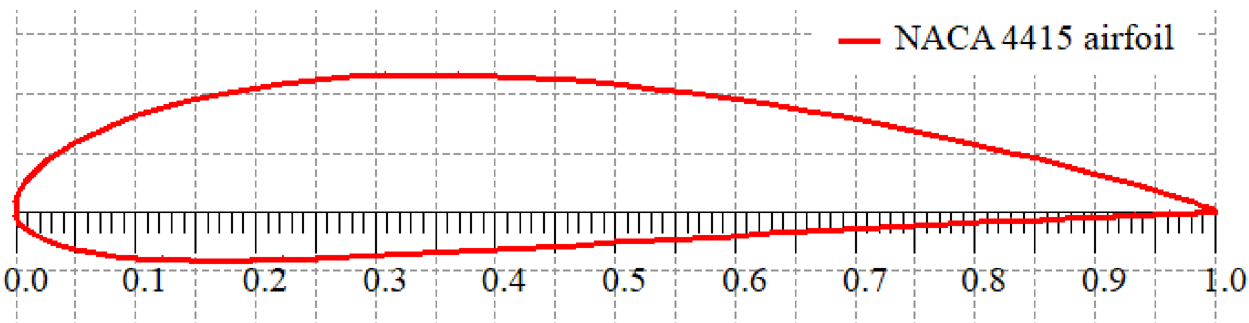

Figure 6: NACA 4415 airfoil

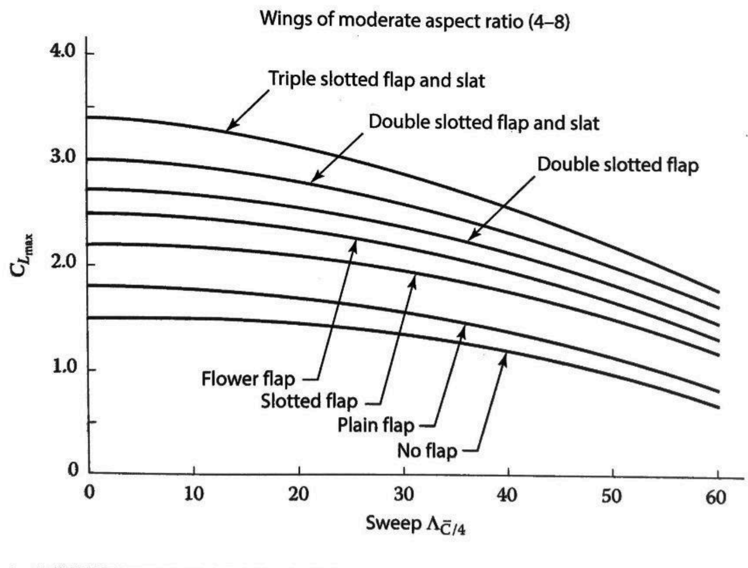

Fig. 5.3 Maximum lift coefficient.

Figure 7: CL, max Performances with different wing configurations

Figure 7. shows how different high lift devices increase the total predicted $\mathbf{C_L}$. With a defined $\mathbf{C_{L,\ max}}$, an estimated take off distance can be calculated using Equation 9. where SG is the initial take off roll resistance. Using triple slotted flaps and leading-edge slats should provide a $\mathbf{C_{L,\ max}}$ value of 3.4. For takeoff, partial flap deflection is typically used to minimize drag. As such, $\mathbf{C_{L,\ Takeoff}}$ was reduced to 70% $\mathbf{C_{L,\ max}}$, such that $\mathbf{C_{L,\ Takeoff}}$ = 2.5.

$$Takeoff\ Distance\ =\ 1.2\times S_G \qquad \text{Eq. 9}$$

The result ultimately came out to 1266 ft. which is an adequate takeoff distance however, the stall speed of the aircraft came out to be 202.5 m/s using Equation 10 which led to a potential issue of the aircraft approaching landing at a dangerous airspeed. To minimize this risk, additional air brake and wheel brake systems must be installed to reduce the landing distance effectively.

$$V_{stall} = \sqrt{\frac{2\times WS}{\rho\times C_{L,max}}} \qquad \text{Eq. 10}$$

5.2 Wing Geometry, Empty Weight, and Fuel Weight

The T-tail conventional Configuration has a wing sizing where the wing is fixed in a conventional rectangular wing. Table 4 showcases all the geometry parameters given to the wing based on the initial wing sizing in CAD.

Table 5: Wing Geometry

Parameter	Value	Units
Reference Area	856.8075	ft^2
Wingspan	82.7919	ft
Mean Aerodynamic Chord	10.3490	ft
Taper Ratio	0.8000	-
Wing Twist	-3	degrees
Wing Position	High	-
Dihedral	2	degrees

The surface area can be derived from Equation 11 which is ultimately the ratio of the final takeoff weight to wing loading. Initially 120 (lbs./ft^2) was chosen as the wing loading value as Table 5.5 correlating the design to a transport jet. This estimation was far too high to meet the takeoff requirement while maintaining a reasonable thrust-to-weight ratio. The wing loading was iterated until it converged to a value of 30 (lbs./ft^2). The wingspan of the aircraft is derived from Equation 12 with the results coming out as 82.7919 ft.

$$S_{ref} = \frac{W_0}{(\frac{W}{S})} \qquad \text{Eq.11}$$

$$b = \sqrt{AR^*\, S_{ref}} \qquad \text{Eq.12}$$

The aircraft is projected to fly with a maximum speed about Mach 0.5. As a result, the aircraft is flying at a low subsonic speed which effectively prevents any shock effect experienced in transonic to supersonic flow (Raymer). Figure 8 demonstrates how the historical data allowed the wings to be designed with no sweep from root to tip of the wing. Additionally, rectangular wings offer the great advantage of having great stall characteristics due to the nature of how the wing will begin stalling from the root of the wing and increase outwards towards midspan (Eagle Pubs). This, in effect, allows the pilot to easily recognize when the aircraft is approaching stall and has far better control during low speeds, making it ideal for a STOL characteristic aircraft for takeoff and landing procedures.

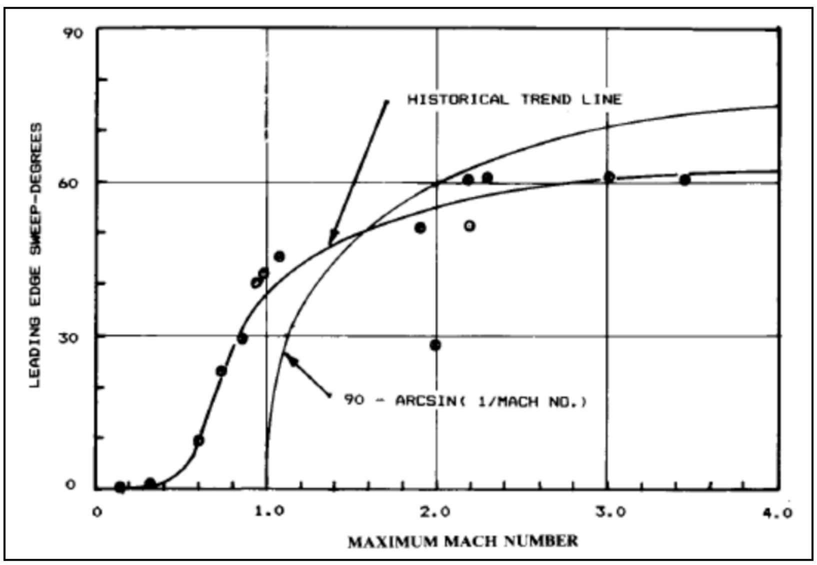

Figure 8: Leading Edge Sweep vs. Maximum Mach Number (Raymer)

Taper ratio would have an overall negative effect on the aircraft STOL performance due to the shrinking nature of the tip relative to the root. While the ideal taper ratio for a rectangular non-swept wing is recommended to be between 0.4-0.5, the team chose to taper the overall wings even less to a 0.80 to further improve stall characteristics. Figure 9 also depicts a relationship between the taper ratio and total induced drag coefficient where having a slight taper would have a smaller reduction of induced drag and overall positive aerodynamic performance (General Aviation Aircraft Design).

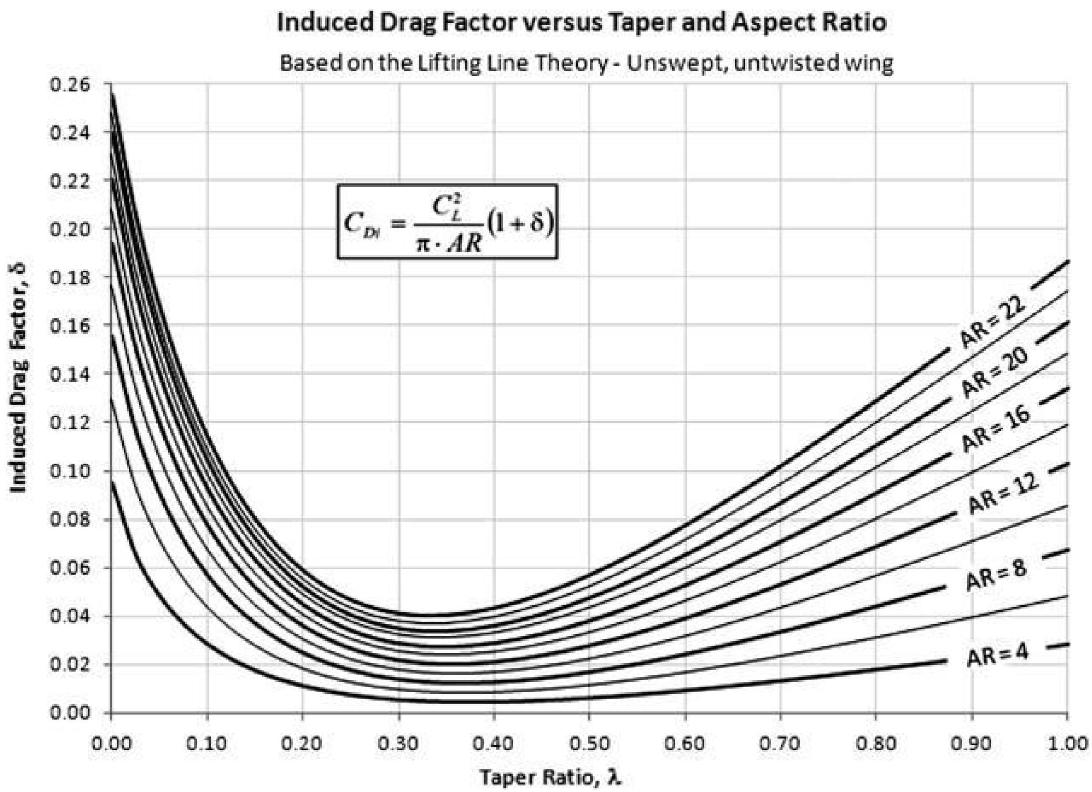

Figure 9: Taper Ratio vs. Induced Drag Graph

The wing will have a -3° washout to improve control at high angles of attack and create a more elliptical lift distribution. This variable was chosen based on historical data for comparable designs (Raymer). Our aircraft design also has a high wing placement, which makes it naturally more stable than a comparable low wing placement aircraft. This property allows for the design to have a relatively shallow 2° dihedral for adequate lateral stability in flight.

5.3 Tail arrangement and Geometry

A T-tail was chosen to keep the horizontal stabilizer out of the jet wash from the engines. This has the added benefit of keeping the tail high above the wing's downwash, leading to improved lift and elevator effectiveness. This may result in better pitch control at higher angles of attack, providing deep stall is avoided through sensible tail placement. T-tails come with the downside of additional structural weight; however, the tail configuration is almost non-negotiable due to the engine placement. A symmetrical NACA 0012 section was chosen for the tail surfaces. This relatively thick section allows for some reduction in structural weight as compared to thinner, lower-drag sections.

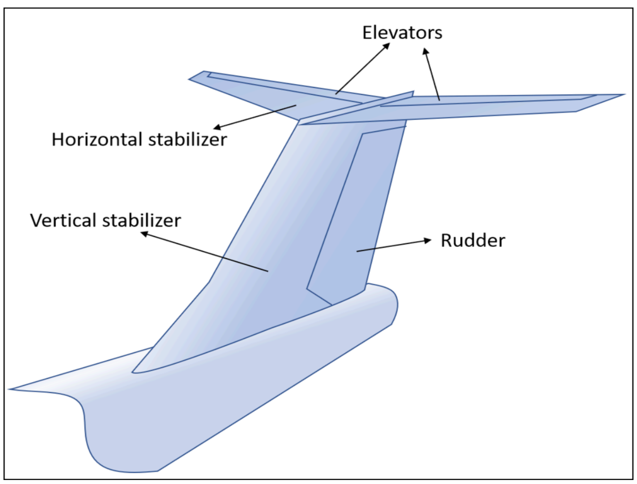

Figure 10: T-Tail Configuration

6. Wing Loading & Thrust-to-Weight Ratio

6.1 Specifications

Takeoff Thrust-to-Weight ratio and Wing Loading can be derived from the Weight Sensitivity Analysis in Section 4. The following equation shows how takeoff thrust-to-weight can be calculated:

$$\left(\frac{T}{W}\right)_{Takeoff} = \left(\frac{T}{W}\right)_{Cruise}\left(\frac{W_{Cruise}}{W_{Takeoff}}\right)\left(\frac{T_{Takeoff}}{T_{Cruise}}\right) \quad \text{Eq.12}$$

Weight fraction between cruise and takeoff stage was already assumed from the initial sensitivity weight estimation (~ 97% of Takeoff Weight) which can be used again for this equation. Thrust-to-weight ratio during cruise can be estimated using Equation 13 which is simply the inverse of the Lift-Drag ratio at cruise stage. The Lift-drag ratio can once again be estimated by Raymer which is about 86.6% of the max Lift-drag ratio.This results in the ratio to be 16.07.

$$\left(\frac{T}{W}\right)_{Cruise} = \frac{1}{\left(\frac{L}{D}\right)_{Cruise}} \quad \text{Eq.13}$$

Finally, for thrust at cruise stage, Eq. 13 can be rearranged to find the thrust at cruise by finding what the weight of the aircraft would be in its cruise stage using the iterated total takeoff weight calculated in Section 4 and the corresponding weight fraction between both stages as shown in Eq. 13b.

$$T_{Cruise} = \frac{W_{Cruise}}{\left(\frac{L}{D}\right)_{Cruise}} \quad \text{Eq.13b}$$

After manually iterating multiple values for take off thrust (starting from 36000 lbf of thrust), the MATLAB code in Appendix A converged at what the team thinks is a reasonable take off thrust value of 23,000 lbf. We reduced the take off thrust with respect to the resulting take off distance shown in Eq. 14 (as the requirement is set at 1200 ft) as solved in Eq. 12 with the take off thrust set at 23,000 lbf. μ was at 0.05. This value is appropriate for representing the rolling resistance of the wheels of the aircraft on a rough surface. The drag to weight ratio was manually calculated to be 0.01.

$$S_{TO} = \frac{{V_{LO}}^2}{2g(\frac{T}{W} - \mu - \frac{D}{W})} \qquad \text{Eq.14}$$

Similarly, the wing loading factor was manually iterated with the take off thrust until the MATLAB script converged to a reasonable take off distance of roughly 1,035 ft, which is well within our STOL requirements.

6.2 Wing Surface Area & Takeoff Thrust

The wing surface area was calculated after the estimated refined take off weight was iterated by MATLAB script. Eq. 15 was rearranged to solve for reference surface area which also equates to the wing surface area in this case. Take off thrust estimations are explained in section 6.1 above.

$$\frac{W_O}{S} = \frac{W_{Takeoff}}{S_{ref}} \qquad \text{Eq.15}$$

6.3 Stall Speed

Aircraft stall speed has been calculated using Equation 10. The stall speed is dependent on wing loading which is discussed in 6.2. Reevaluating stall speed to better match the initial aircraft takeoff distance parameter gives a value of 123 ft/s.

7. Initial Sizing

7.1 Refined Weights

Solving our variables in Section 6 allows us to calculate a new refined total takeoff, empty, and fuel weight estimations using a more in-depth formula. A new iteration formula to find total takeoff weight can be described with Equation 16.

$$W_O = W_{Crew} + W_{Fixed\ Payload} + W_{Dropped\ Payload} + W_{Fuel} + (\frac{W_E}{W_O})W_O \qquad \text{Eq.16}$$

Within the new equation, the only extra component that is introduced is the dropped payload weight which will be reduced to zero as all payload and crew weight will remain the same throughout the full duration of flight. Empty to Takeoff Weight Fraction is another component that was calculated using the equation found in Raymer Table 6.1. For our project purposes, all of the coefficients utilized in the equation will be modelled after the "Jet Transport" configuration.

$$W_e/W_0 = [a + b\, W_0^{C1} A^{C2} (T/W_0)^{C3} (W_0/S)^{C4}\, M_{max}^{C5}] K_{vs}$$

fps units	a	B	C1	C2	C3	C4	C5
Jet trainer	0	4.28	−0.10	0.10	0.20	−0.24	0.11
Jet fighter	−0.02	2.16	−0.10	0.20	0.04	−0.10	0.08
Military cargo/bomber	0.07	1.71	−0.10	0.10	0.06	−0.10	0.05
Jet transport	0.32	0.66	−0.13	0.30	0.06	−0.05	0.05

K_{vs} = variable sweep constant = 1.04 if variable sweep and 1.00 if fixed sweep

Figure 11. Empty -Takeoff Weight Fraction Equation

Using the takeoff thrust-to-weight ratio, wing loading, Aspect Ratio, and Max Mach speed, $\frac{W_e}{W_0}$ can be calculated with each iteration of total takeoff weight. Fuel Weight is another component that needs to be iterated as well. Section 3 already has a summed weight fraction of each stage of the flight mission even with the weight dropped payload being negligible. As a result, our team chose to keep the previous weight fraction estimation to calculate our fuel weight. Once the equations are set to run the iterations, our team chose 12500 lbs as the initial takeoff weight estimate as well as 0.01 for the tolerance. The updated weights values are shown in the following Table 6.

Table 6. Refined Weights

Weight Class	New Total Weight (lbs.)
Payload	330
Crew	370
Take-off	37990.59
Empty Aircraft	25463.95
Dropped Payload	0
Fuel	152526.63

7.2 Geometry Sizing

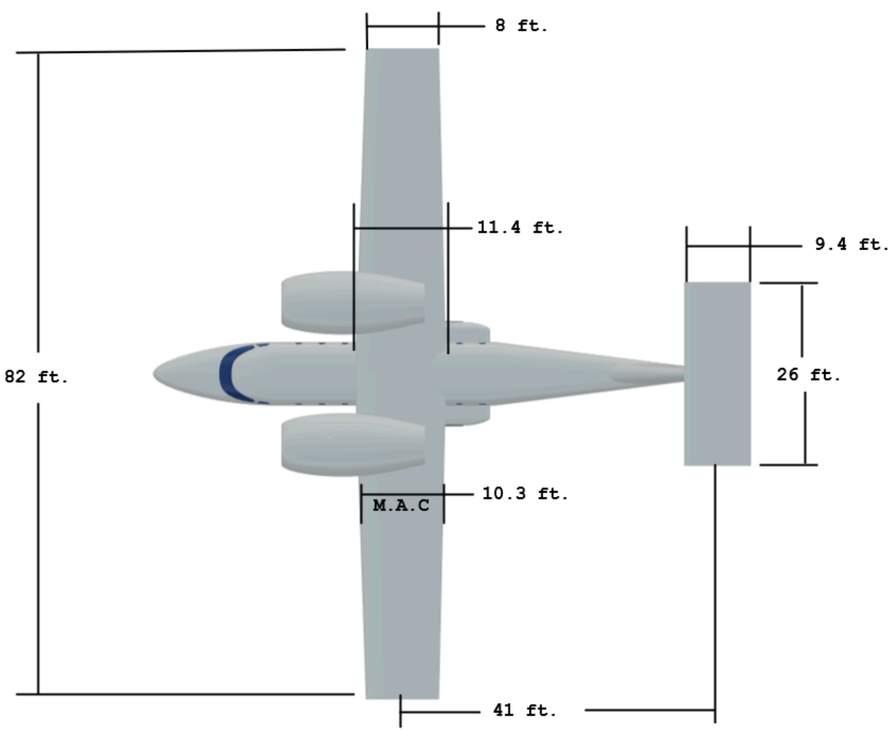

Figure 12: Wing and Tail Sizing

Table 7. Wing and Tail Parameters

Parameter	Value	Units
Reference Area	844.2353	ft^2
Wingspan	82.1820	ft
Mean Aerodynamic Chord	10.2728	ft
Taper Ratio	0.8000	-
Wing Twist	-3	degrees
Wing Position	High	-
Dihedral	2	degrees

Figures 12 and Table 7 provide a detailed summary of our airplane's aerodynamic surface sizing and geometry. Our wing reference area was determined through our iterative sizing in MATLAB, with aspect ratio and tail volume being chosen based on historical design data from Raymer. A horizontal tail volume of 0.9 was selected, which is slightly larger than the 0.8 recommended for jet transports. This is to allow for a wider CG range and large pitching moments created by the engines and flaps.

7.3 Control Surfaces

Ailerons & Flaps

Our ailerons size can be determined using historical guidelines from Raymer. With the aileron chord being about 2.4 ft. while the wing's mean aerodynamic chord is 10.3. The ratio, 0.23, can be used in Raymer's Figure 6.3 where the percentage of aileron span to total wingspan can be estimated flight. The team chose 16ft. as the aileron control surfaces length would be around 39% of the wingspan. Flaps are also installed alongside the ailerons which will have the same length but a higher chord length of 3ft. to yield higher surface area and lift performance. Leading-edge slats are placed outboard of the engines, and they measure 26ft. in length.

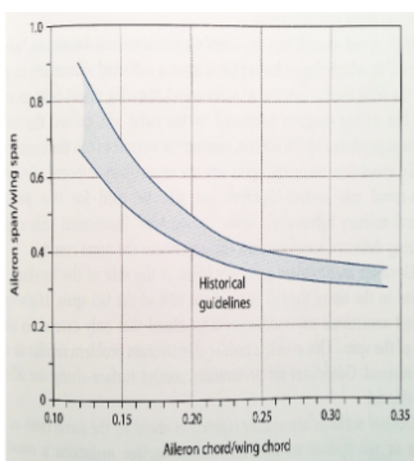

Figure 13: Percentage of Aileron Span vs. Aileron-Wing Chord Ratio

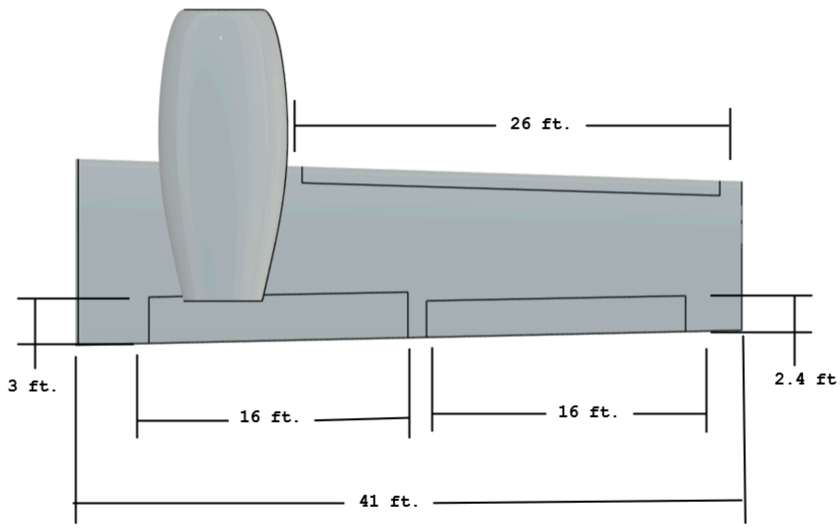

Figure 14: Wing control surface dimensions

Elevator & Rudder

The team decided that the elevator should be approximately 37% the chord of the horizontal stabilizer at 3.5 ft and 20 ft in span. Although Raymer's table 6.5 was suggesting the sizing to be around 32%, the high mounted engines of the configuration necessitated the bigger elevator size to counteract the pitching moment. The rudder was sized to be 30% the chord of the vertical stabilizer and spanned about 12 ft. This sizing seems to be on par with what the value is shown in Raymer's table 6.5.

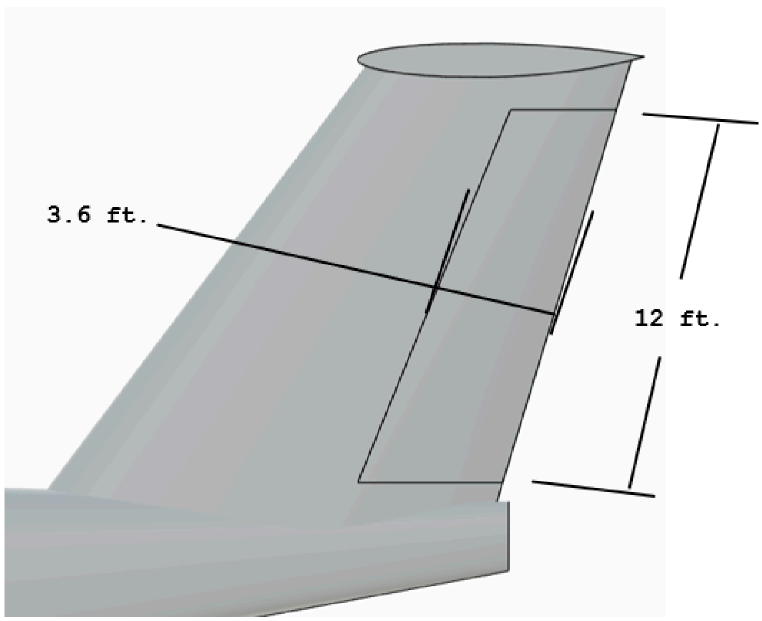

Figure 15: Vertical Stabilizer Dimensions

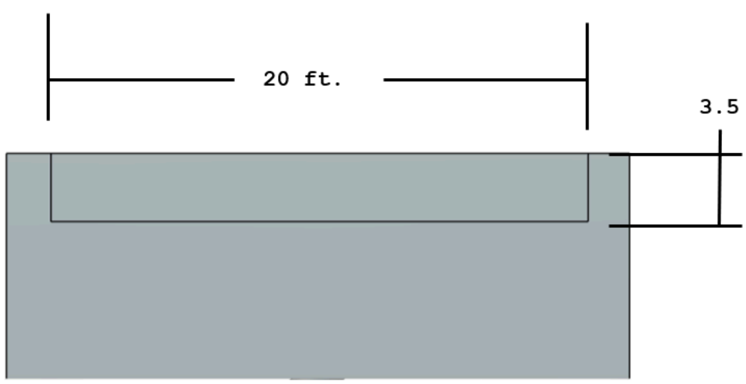

Figure 16: Horizontal Stabilizer Dimensions

8. Configuration Layout & Loft

The finalized sizing for all components of the aircraft allows a more refined 3D model. Loft commands were utilized to transform 2D airfoil profiles of the wings into a full extruded wing piece that fits with the wingspan and wing area requirement. Similarly, fuselage cross-sections were lofted at the nose and tail to create an aerodynamic shape. A constant cross-section was used for the cabin for manufacturability.

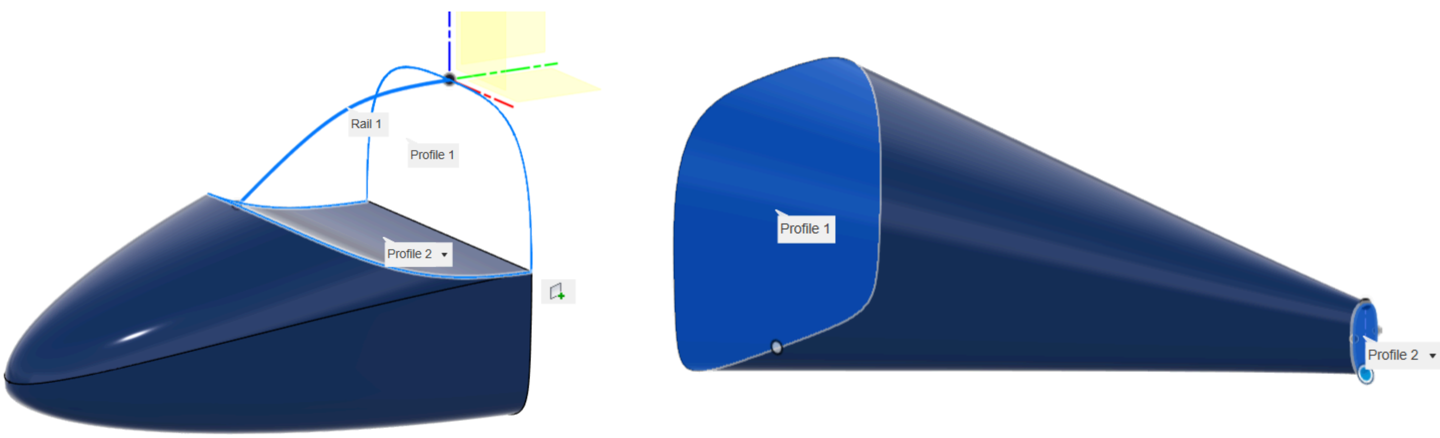

Figure 17: Nose and Tail Cone Lofts

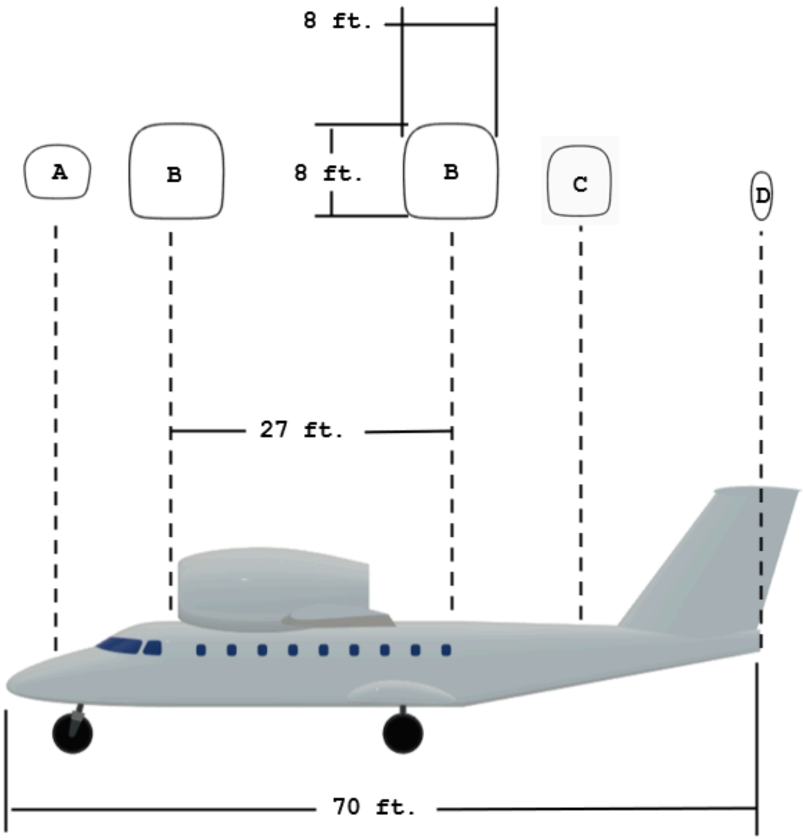

Figure 18: Cross Section Profiles of Fuselage Across Fuselage Length

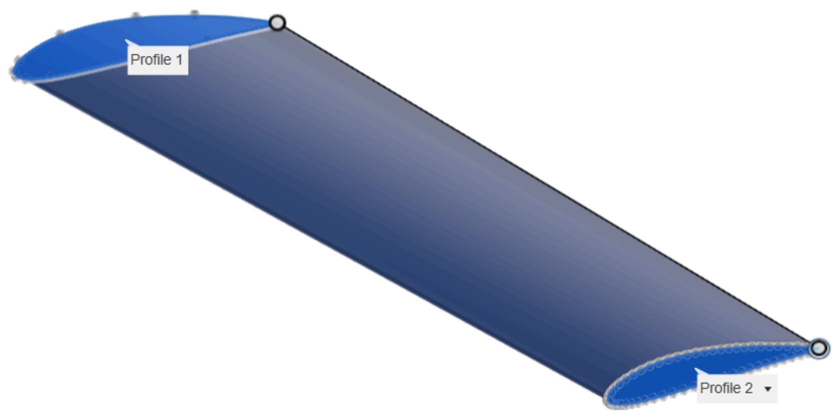

Figure 19: Wing loft

9. Special Considerations in Configuration Layout

9.1 Aerodynamic Considerations

Wing design

The mission of this aircraft is to achieve short takeoff and landing. This requires a wing capable of producing a high maximum lift coefficient. To meet this requirement, the plane will use a triple slotted flap and slat configuration. This system will increase lift at low speeds and will increase the wing camber and extend the wings surface area. Using slats helps maintain attached airflow over the wing at high angles of attack. This will reduce stall speed and improve lift at low speeds. However, having the triple slotted flap and slat configuration has several trade-offs. The added complexity of the design will increase the overall weight. It will also increase the aircrafts drag when the flaps and slats are deployed. The increased drag will decrease cruise efficiency and the deployed slats and flaps will need to be retracted after take off to minimize drag. The increased drag of the slats and flaps when deployed actually can be used to benefit the mission during landing. It will help slow down the aircraft quickly and shorten the landing distance.

Our wings on our aircraft do not have any sweep. The sweep would add unnecessary complexity. Also would decrease lift performance for the short takeoff and landing capabilities of the plane.

Fuselage

The fuselage plays a critical role in the aircraft's aerodynamic performance. One of the primary metrics used in aircraft performance is S_wet. S_wet is the total surface area exposed to airflow. Having a high S_wet will increase skin friction drag on the aircraft causing parasitic drag. Our aircraft has an S_wet value of 4781 squared ft. This is right around near to what is expected for subsonic aircrafts. Our aim was to reduce S_wet by designing a compact internal layout to lower volume. Our group also focused on streamlining the fuselage to get rid of unnecessary protrusions on the aircraft. Focusing on these reduces S_wet while also lowering the aircrafts overall weight. Having a poor fuselage design can lead to excessive flow separation which will increase drag and induce instability for the aircraft. To mitigate these effects the fuselage was designed with smooth longitudinal control lines, adding fillets to where the wing and the fuselage meetaking sure the forebody is streamlined along with the base area is tapered to reduce low pressure wake zones.

The fineness ratio of the fuselage plays a big role in aerodynamic efficiency. The length of our plane is 71 feet and the maximum diameter is 8 feet. Our aircraft has a fineness ratio of 8.875. While this is slightly larger than the average of 6-8, this should not increase drag by a significant amount due to the shallow slope of the fineness ratio curve in this range.

The shape of the fuselage cross section can also significantly impact drag and internal volume . Our team is going with a square cross section to efficiently use the internal space. The square shape makes the aircraft simpler to produce and the filleted corners reduce drag.

Cleanliness

Our aircraft will always try to be as aerodynamically clean as possible, within the limits of aluminum construction. As such, gaps and rivets are minimized wherever possible. Retractable landing gear will be implemented, and fairings will cover all external devices such as antennas or sensors.

9.2 Structural Considerations

The purpose of our mission is to provide a design for an aircraft that can support short takeoff and landing capabilities. The structural design needs to prioritize strength, efficiency to achieve our goal. The fuselage will use an aluminum frame and longeron system. The frame handles most of the radial stresses while the longerons run longitudinally to prevent fuselage bending along its length, with a skin taking torsional loads. The wings are built around front and rear spars designed to carry lift and take on bending loads. Ribs support the skin, defininging the airfoil shape. These spars are involved in a wingbox carry through structure that spans the fuselage and forms a continuous load path while minimizing fuselage weight. The wing box is tied into the fuselage to add stiffness and safety for crashworthiness. Flutter is considered as well, and mitigated by mass balanced control surfaces and high torsional rigidity in the wings.

10. Crew and Passenger Station

10.1 Crew Station

The cockpit of the aircraft accounts for the 2 crew members where the seating arrangement and dimensions was referenced to the Twin Otter's cockpit interior shown in Figure X. The height of the seat being Xin. fits to the general 5th to 95th percentile height range for males and 20th percentile range for females. In addition to the seat sizing, an adequate over nose angle was calculated based on the approach angle of attack and approach speed assuming speed is measured in knots (Eq X).

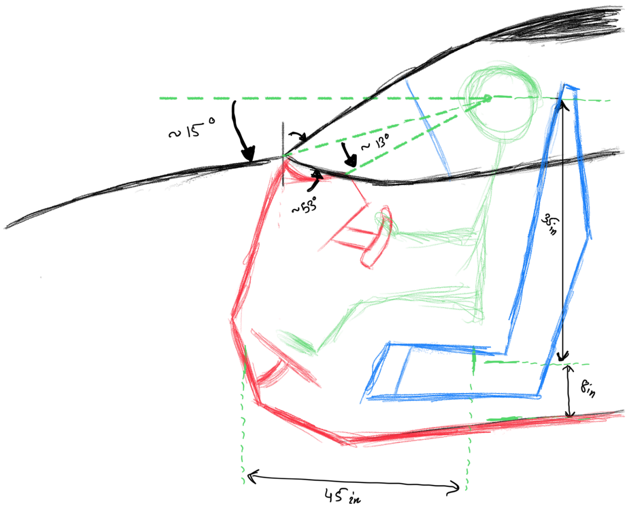

Figure 20: Cockpit Seat Sizing

$$\alpha_{overnose} = \alpha_{approach} + 0.07\, V_{approach} \quad \text{Eq.17}$$

10.2 Passenger Area

The passenger compartments will have 18 seats where each side of the fuselage will have one seat besides the window as shown in the floor map in Fig.21. All of the seats will be forward facing and the bathroom is placed and accommodated in the back of the aircraft as well as a pantry cabinet for snack and drinks storage. Note that the bathroom and pantry's weight is accounted for in the empty aircraft weight.

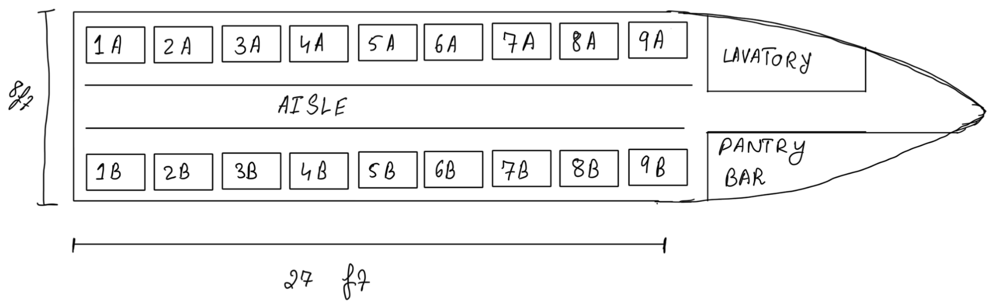

Figure 21: Passenger Area Seating Chart

Within the cross section of the passenger seats, Fig.X showcases further dimensions of the seats, aisle and head room. All passengers will be provided with First Class upholstered leather armchairs with a sufficient legroom space of 3ft and capable of recline.

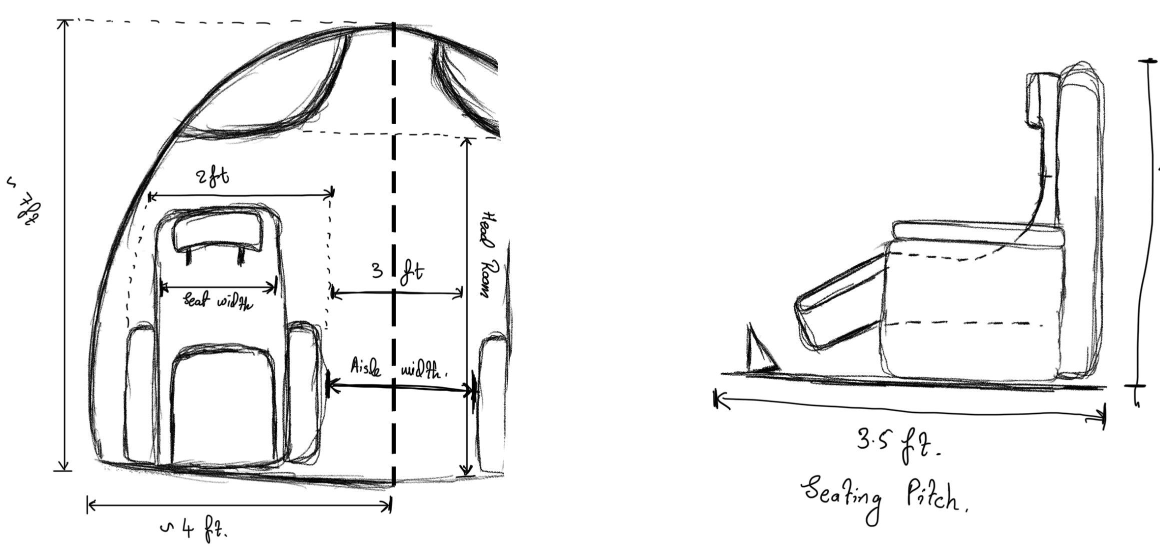

Figure 22: Passenger Seat Sizing

11. Propulsion and Fuel System Integration

11.1 Propulsion System

The high bypass jet turbofan is the ideal propulsion configuration for our aircraft as it offers high performance efficiency at subsonic speeds and offers quieter noise than other engines. To calculate the bypass ratio of the turbofan, an average of high bypass ratio.from most modern private jets were used as baseline for the sizing calculations. In terms of engine sizing for the turbofan, the engines' dimensions such as length, diameter, and weight can be calculated using Eq. 18 - 20 below.

The required length for the engine was calculated to be about 17.47 ft. at minimum, and the required diameter needs to be about 4.32 ft. The weight of the engine was calculated to be approximately 1964.86 lbs. These numbers were calculated based on the takeoff thrust (T) at 11,500 lbs from each engine and a Mach number of 0.55 with a BPR estimated at 5.

The fuel of the aircraft will be stored within the wing structure inside rubber bladders as it mitigates the risk of the fuel seeping into the fuselage and the main cabin. Furthermore, rubber bladders are also self sealing in case of wing strikes, which can help reduce the fire hazard and critical system failures due to it.

$$L = 0.185T^{0.4}M^{0.2} \qquad \text{Eq.18}$$

$$D = 0.033T^{0.5}e^{(0.04BPR)} \qquad \text{Eq.19}$$

$$W = 0.084T^{1.1}e^{(-0.045BPR)} \qquad \text{Eq.20}$$

12. Landing Gear and Subsystems

The aircraft systems are specifically designed to support off-airport operations. As such, the aircraft will use retractable tricycle landing gear, featuring oversize tundra tires for operation on rough terrain. Due to the narrow fuselage and high wing, the main gear must be stowed in pods on the sides of the fuselage, to widen the stance on the ground and provide aerodynamic fairings for the mechanism when stowed. The nose gear retracts into the nose. While retractable gear on STOL aircraft is unconventional, it is required to meet the speed and range requirements of our aircraft. That's not to say this gear configuration is unique - it's widely seen on high-wing retractable gear aircraft such as the C-130 and An-72.

The landing gear and flight controls are actuated using tandem hydraulic systems. This allows for improved reliability in the case of component failure. Two actuators are used per control surface, such that only a failure of both hydraulic systems simultaneously could result in a loss of control.

13. Safety

13.1 Crash Safety

Airplanes are one of the safest modes of transportations. For every 100 million miles traveled via car your risk of dying is .57%. However, in an airplane this decreases to only .003% every 100 million miles. In the event of a survivable plane crash there are precautions that passengers can take to increase survivability. Survivable plane crashes examples could be an airplane skidding across the runway during takeoff. If you are in the unfortunate event of being in a plane at full speed and it's going to crash, sit in whatever position is most comfortable. Airplanes have a limited amount of space and certain weight that has to be maintained so some safety during a crash falls upon the passenger of the airplane. Brace positions will be provided to passengers so passengers will avoid throwing themselves into walls or the seat in front of them. The FAA has regulations on seat positions and constantly changes to prevent injuries. Airplanes themselves already do a good job at taking some of the load the plane would put out in a crash however force will no matter what goes to the passenger so they must be prepared. Below in Fig. 23, some of the recommended seating positions the FAA finds out is best in the event of a crash.

Figure 23: Brace Positions in forward facing seats

These positions should be used in the event of the crash because it expects the forward force the passenger will experience and cause the passenger to transfer the energy through them and into the seat in front of them through their forearms instead of their face. Having your arms like this acts like an airbag and may mitigate major injuries to minor injuries.

13.2 Cabin Safety

Our aircraft implements many safety features to safely transport passengers. Our aircraft will follow FAA guidelines to make sure our aircraft holds up to the cabin safety index. Our cabin will be pressurized at high altitudes to make sure the air is not too thin to breathe safely. We will also provide access to oxygen masks if the cabin pressurization ever fails. Along with our air systems we will make sure HEPA filters are implemented to remove bacteria in the cabin air to enjoy a comfortable transportation service. In the event of an emergency fire safety systems will be put in place such as smoke detectors. Fire extinguishers will be put into easy to reach areas and plane cabins will have material choices that are not flammable. Seat belts will also be required to be worn by the passengers during takeoff, landing and turbulence to ensure safety. All of our safety features and emergency exits and life vests will be easily found in the safety brochures located in front of every passenger seat. All crew members will also be trained in first aid and evacuation procedures to ensure if an emergency occurs they can efficiently guide the people.

13.2 General airline Safety

Our aircraft undergoes normal maintenance checks after every landing to check system functionality and make sure no damage has occurred on the aircraft. Our pilots go through extensive training and have to undergo simulator flights and in flight training to handle emergency situations. The luggage in the cabin is put away and closed to ensure no luggage falls on any passengers. The communication systems of the aircraft also allows continuous contact with air traffic control and the flight has navigation systems such as radar, GPS and collision avoidance.

14. Applicable Design Standards

Our Aircraft has constraints set for standards by multiple organizations. The FAA, Army-Navy, National Aerospace Standard hardware and the society of automotive engineers (SAE) all have their own aerospace related specifications for aircraft. Our aircraft follows regulations by several organizations and this section will cover some standards our airplane follows. According to the FAA Federal Aviation Regulations (FAR) Part 25 shows fundamental criteria for commercial aircrafts and that every major component of the aircraft must show compliance from simulations, analysis and from physical testing. FAA also has FAR 25.1301-25.1329 that influence our cockpit layout making sure pilots have readable signage and control reachability and certain seating layouts for pilots. All the materials used on our flight also follow standards set by SAE AMS that set the specifications for some materials like composites and aluminum. ASME also sets Boiler and Pressure Vessel Code that is used in our design to set pressure vessel standards. Pressure Vessel standards ensure that oxygen tanks and hydraulics in the aircraft dont leak and are inspected regularly. The aircraft accountants make sure standards are set to meet safety, legal and engineering expectations of all aircrafts to enhance the aircrafts safety and reliability.

15. Cost Analysis based on DAPCA IV

The modified DAPCA IV model was used to calculate the production cost of 500 aircrafts. The model includes engineering, tooling, manufacturing, and quality control labor hours, multiplied by the respective hourly rates.The cost of manufacturing materials, procuring raw resources, and purchasing hardware are also factored into the total cost. The engine production costs are calculated separately, and the team assumed a turbojet engine with total thrust of 13000 lb. Avionics costs are based on concurrent values. The final cost combines all the previously mentioned expenses, RDT&T and flyaway expenses. A cost breakdown of each sector is provided in table X. The final result is $ 8.724 billion for 500 units and $ 17.448 million for a single unit to be produced. Eq. X was used to perform the final calculation.

Expense	Cost ($)
Engineering	1612800000
Tooling	1082800000
Quality Control	204170000
Development Support	205050000
Flight Test	46532000
Manufacturing Material	1848400000
Engine Production	1818700
Avionics	200000000
Manufacturing	1705600000
Total Cost	8724000000

$$RDT\&E + flyaway = H_E R_E + H_T R_T + H_M R_M + H_Q R_Q + C_D + C_F + C_M + C_{eng} C_{eng} + C_{avionics}$$

16. How does an airplane fly:

In the simplest of terms, airplanes fly forward and upward by pushing air backward and downward. If the following conditions are met, the airplane will fly.

Lift: As a wing moves through the air, the air is directed around the wing such that the pressure distribution around the wing has a net-upwards force component. This is typically achieved using a cambered (curved) airfoil shaped acting with a positive angle of attack.

Thrust: Engines push the air backward, which pushes the airplane forward. Every type of airplane, whether it uses a jet, rocket, or propeller, uses momentum exchange with the air (or propellant) to create thrust. Gliders are the sole exception, which use a combination of gravity and rising air currents to gain energy.

Stability: Oftentimes, a tail is needed to maintain the airplane's attitude as required for flight. A tail consists of aerodynamic surfaces, often with control surfaces, that naturally create a restoring force when the angle of attack deviates from the trim angle of attack. However, certain airfoils and planforms can achieve this stable condition without additional surfaces, leading to stable tailless configurations such as the flying wing and plank. Unstable configurations can be made stable with modern flight controls, which offer maneuverability and efficiency benefits.

Appendix A - MATLAB Code (EXAM 1 estimations)

```
%% MAE434 - Aircraft Design Exam 1 (Matlab Script)
% Team: Charlie
% Date: 3/11/2025
%-----------------------------------------------------------------------------
%% Project Script Starts Here
clear all; close all; clc;
%-----------------------------------------------------------------------------
%% Input Variables
alt_ft = 40000; %ft (cruise altitude)
alt_m = convlength(alt_ft,'ft','m'); %Cruise altitude in meters
[alt_T, alt_a, P, alt_rho] = atmosisa(alt_m);
rho_imperial = alt_rho/515.4;
%velocity calculations
V_cruise = 515; %ft/s
alt_a_fts = alt_a*3.281; %ft/s
M_cruise = V_cruise/alt_a_fts; %mach number
fprintf('<strong>Initial Velocity Calculations :)</strong>\n')
fprintf('Cruise Velocity: <strong>%5.2f</strong> ft/s\n',V_cruise)
fprintf('Speed of Sound at 40,000 ft: <strong>%5.2f</strong> ft/s\n',alt_a_fts)
fprintf('Cruise Mach Number: <strong>%5.2f</strong>\n',M_cruise)
fprintf('------------------------------------------------------------\n')
%Plane/Aerodynamics/Payload specs
w_pl = 185*18; %lbs (taken from the De Havilland Aircraft - Twin Otter)
w_crew = 185*2; %lbs (avg passenger mass = 185, we plan to carry 18 passengers)
AR = 8; %Estimated Aspect Ratio
swet_sref = 4.34; % 6.2 and & 5 %CAD Reference surface area ratio
```

```
K_ld = 15.5; %Civilian Jets ---> Same Page as table 3.6 Aircraft Design - Raymer
fprintf('<strong>Engine Aircraft/Aerodynamics/Payload Specifications :)</strong>\n')
fprintf('Payload Weight: <strong>%5.2f</strong> lb\n',w_pl)
fprintf('Crew weight: <strong>%5.2f</strong> lb\n',w_crew)
fprintf('Aspect Ratio: <strong>%5.2f</strong>\n',AR)
fprintf('Reference Surface Area Ratio: <strong>%5.2f</strong>\n',swet_sref)
fprintf('---------------------------------------------------------------\n')
%range and loitering
Range = 3000 * 6076; %we're planning to fly around 3000 Nautical Miles
L_time = 45*60; %45 min loiter time, converted to seconds
fprintf('<strong>Engine Properties - High-bypass turbofan :)</strong>\n')
fprintf('Range: <strong>%5.2f</strong> ft\n',Range)
fprintf('Loiter time: <strong>%5.2f</strong> seconds\n',L_time)
fprintf('---------------------------------------------------------------\n')
%Engine Properties - High-bypass turbofan
C_cruise = 0.5/3600; %1/s %SFC ----> Table 3.3 Aircraft Design - Raymer
C_loiter = 0.4/3600; %1/s %SFC ----> Table 3.3 Aircraft Design - Raymer
fprintf('<strong>Engine Properties - High-bypass turbofan :)</strong>\n')
fprintf('C_loiter: <strong>%e</strong>\n',C_loiter)
fprintf('C_cruise: <strong>%e</strong>\n',C_cruise)
fprintf('---------------------------------------------------------------\n')
%Solving for characteristics
L_D_max = K_ld*sqrt(AR/swet_sref); %loiter l/d ratio
L_D_cruise = 0.866*L_D_max; %cruise l/d ratio for jet
fprintf('<strong>5-Stage Fuel & Weight Fraction Calculation :)</strong>\n')
fprintf('L/D (MAX): <strong>%5.2f</strong>\n',L_D_max)
fprintf('L/D (CRUISE): <strong>%5.2f</strong>\n',L_D_cruise)
```

```
fprintf('--------------------------------------------------------------\n')
%5-stage fuel characteristics
w_10 = 0.97; %Take off
w_21 = 0.985; %Climb
w_32 = exp(-(Range*C_cruise)/(V_cruise*L_D_cruise)); %cruise
w_43 = exp(-(L_time*C_loiter)/(L_D_max)); %Loiter
w_54 = 0.995; %landing
w_50 = w_10*w_21*w_32*w_43*w_54; %total weight decrease ratio
w_f0 = 1.06*(1-w_50); %fuel to weight ratio -----> 6% residual weight allowance
fprintf('<strong>5-Stage Fuel & Weight Fraction Calculation :)</strong>\n')
fprintf('weight fraction during takeoff: <strong>%5.4f</strong>\n',w_10)
fprintf('weight fraction during climb: <strong>%5.4f</strong>\n',w_21)
fprintf('weight fraction during cruise: <strong>%5.4f</strong>\n',w_32)
fprintf('weight fraction during loiter: <strong>%5.4f</strong>\n',w_43)
fprintf('weight fraction during landing: <strong>%5.4f</strong>\n',w_54)
fprintf('weight fraction (total weight decrease ratio): <strong>%5.4f</strong>\n',w_50)
fprintf('weight fraction (fuel with 6-percent residual weight allowance): <strong>%5.4f</strong>\n',w_f0)
fprintf('--------------------------------------------------------------\n')
%--------------------------------------------------------------------------
%% determining takeoff weight (Sensitivity analysis)
A_val = 1.02; %table 3.1
C_val = -0.06; %table 3.1
K_vs = 1.0; %constant sweep
guess_to_weight = 12500; %lb %initial weight guess from the Twin Otter
rhs = 1;
tolerance = abs((1 - (guess_to_weight/rhs))*100); %tolerance condition
iteration = 0;
```

```
%while loop is going to iterate a solution for the empty weight of the
%aircraft
while tolerance > 0.01
   guess_to_weight = guess_to_weight + 0.1; %increasing weight by 5lb
    w_e0  = A_val*(guess_to_weight^C_val)*K_vs; %estimating empty weight ratio based on the initial guess
    rhs  =  (w_pl  +  w_crew)/(1  -  w_f0  -  w_e0); %updating the right hand of the equation
     tolerance  =  abs((1  -  (guess_to_weight/rhs))*100);  %tolerance  condition updated
   iteration = iteration + 1;
end
%weight calculations (empty and fuel)
w_f = rhs*w_f0;
w_e = rhs - w_f;
fprintf('<strong>Weight Calculation Results below :)</strong>\n')
fprintf('The  Total  Take-off  Weight  of  the  aircraft  is  estimated  at: <strong>%5.2f</strong> lbs\n',guess_to_weight)
fprintf('The   Fuel   Weight   of   the   aircraft   is   estimated   at: <strong>%5.2f</strong> lbs\n',w_f)
fprintf('The   Empty   Weight   of   the   aircraft   is   estimated   at: <strong>%5.2f</strong> lbs\n',w_e)
fprintf('---------------------------------------------------------------\n')
%---------------------------------------------------------------------------
%% Wing Design and Parameters
WS = 30; %lb/ft^2 %table 5.5 transport jet value (we had to reduce it to reduce the take off distance and also that 120 from the book was not reasonable as if was of a full-scale bomber)
taper_ratio = 0.8; %estimated from the CAD Design
s_ref = rhs/WS; %reference are
wing_span = sqrt(AR*s_ref); %wingspan
```

```
%root_chord = (2*s_ref)/(wing_span*(1-taper_ratio)); %optimal root chord
%tip_chord = taper_ratio*root_chord; % optimal tip chord
%mean_chord = ((2/3)*root_chord*(1+taper_ratio+taper_ratio^2))/(1+taper_ratio);
mean_chord = s_ref/wing_span;
wing_area = s_ref;
cl_cruise = (2*guess_to_weight)/(rho_imperial*V_cruise^2*wing_area);
s_wet = swet_sref*s_ref;
fprintf('<strong>Wing Area Calculation ASSUMING Rectangular Wings  Results below :)</strong>\n')
fprintf('Estimated Coefficient of Lift (cruise): <strong>%8.4f</strong>\n',cl_cruise)
fprintf('The reference area of the wings is estimated at: <strong>%8.4f</strong> ft^2\n',s_ref)
fprintf('Estimated Wetted Surface Area of the wing: <strong>%8.4f</strong> ft^2\n',s_wet)
fprintf('Estimated wing span of the aircraft is: <strong>%8.4f</strong> ft\n',wing_span)
fprintf('Estimated mean aerodynamic chord of the wings is: <strong>%8.4f</strong> ft\n',mean_chord)
fprintf('Estimated Taper Ratio (Not Applied) of the wings is: <strong>%8.4f</strong>\n',taper_ratio)
fprintf('---------------------------------------------------------------\n')
%---------------------------------------------------------------------------
%% Take-off distance calculator
%%SG - initial take off roll, before rotation
TW = 1.0; %minimizing takeoff distance
CL_to = 0.5;
CL_max = 2.5; %jet flaps/slats/flaps/
CD0 = 0.03;
e = 0.8;
K = 1/(pi*e*AR); %K-factor from Raymer takeoff distance eqns
```

```
vstall = sqrt(2*WS/(rho_imperial*CL_max));

Vto = 1.2*vstall;

%from raymer pg 673

mu = 0.05; %rolling resistance

KT = TW - mu;

KA = rho_imperial / ((2*WS))*(mu*(CL_to)-(CD0)-(K*CL_to));

SG = (1/(2*32.2*KA))*log((KT+KA*Vto^2)/(KT));

TakeOffDistance = SG*1.3;

fprintf('<strong>Takeoff distance calculation :)</strong>\n')

fprintf('Estimated stall speed: <strong>%8.4f</strong> ft/s\n',vstall)

fprintf('Estimated        takeoff        distance:        <strong>%8.4f</strong>
ft\n',TakeOffDistance)

fprintf('---------------------------------------------------------------\n')

fprintf('<strong>Script Ends Here As Of Now</strong>\n')

%% Script Ends here as of now.
```

Appendix B - Sample Calculation

Input Variables:

$$alt\ ft = 40{,}000\ ft\ or\ 12{,}192\ m$$

$$V_{cruise} = 515\ \text{ft/s}$$

$$alt_a = 295.0696\frac{m}{s}$$

$$alt_{a_{\frac{ft}{s}}} = (alt_a)(3.281) = 968.1234\frac{ft}{s}$$

$$M_{cruise} = \frac{V_{cruise}}{alt_{a_{\frac{ft}{s}}}} = .5320$$

Plane/Aerodynamics/Payload Properties

$$w_{PL} = 4300$$

$$w_c = 3515$$

$$AR = 8$$

$$\frac{S_{wet}}{S_{ref}} = 6.2$$

$$k_{ld} = 15.5$$

Range and loiter Estimates

$$Range = (3000)(6076)$$

$$Loiter\ Time = (45)(60)$$

Engine Properties -> High Bypass Turbo fan

$$C_{cruise} = \frac{0.5}{3600}$$

$$C_{loiter} = \frac{0.4}{3600}$$

$$\left(\frac{L}{D}\right)\max = (k_{ld})\left(\sqrt{\frac{AR}{\frac{S_{wet}}{S_{ref}}}}\right) = 21.0442$$

$$\left(\frac{L}{D}\right)cruise = 0.866\left(\frac{L}{D}\right)\max = 18.243$$

Appendix C - MATLAB Code (Exam 2 - Refined Estimations)

```
%% MAE434 - Aircraft Design Exam 2
% Team: Charlie
% Date: 3/11/2025
%--------------------------------------------------------------------------
%% Project Script Starts Here
clear all; close all; clc;
%--------------------------------------------------------------------------
%% Input Variables
alt_ft = 40000; %ft (cruise altitude)
alt_m = convlength(alt_ft,'ft','m'); %Cruise altitude in meters
[alt_T, alt_a, P, alt_rho] = atmosisa(alt_m);
rho_imperial = alt_rho/515.4;
%velocity calculations
V_cruise = 515; %ft/s
alt_a_fts = alt_a*3.281; %ft/s
M_cruise = V_cruise/alt_a_fts; %mach number
fprintf('<strong>Initial Velocity Calculations :)</strong>\n')
fprintf('Cruise Velocity: <strong>%5.2f</strong> ft/s\n',V_cruise)
fprintf('Speed of Sound at 40,000 ft: <strong>%5.2f</strong> ft/s\n',alt_a_fts)
fprintf('Cruise Mach Number: <strong>%5.2f</strong>\n',M_cruise)
fprintf('-------------------------------------------------------------\n')
%Plane/Aerodynamics/Payload specs
w_pl = 185*18; %lbs (taken from the De Havilland Aircraft - Twin Otter)
w_crew = 185*2; %lbs (avg pssnger mass = 185, we plan to carry 18 passangers)
AR = 8; %Estimated Aspect Ratio
swet_sref = 4.34; % 6.2 and & 5 %CAD Reference surface area ratio
```

```
K_ld = 15.5; %Civilian Jets ---> Same Page as table 3.6 Aircraft Design - Raymer

fprintf('<strong>Engine Aircraft/Aerodynamics/Payload Specifications :)</strong>\n')

fprintf('Payload Weight: <strong>%5.2f</strong> lb\n',w_pl)

fprintf('Crew weight: <strong>%5.2f</strong> lb\n',w_crew)

fprintf('Aspect Ratio: <strong>%5.2f</strong>\n',AR)

fprintf('Reference Surface Area Ratio: <strong>%5.2f</strong>\n',swet_sref)

fprintf('--------------------------------------------------------------\n')

%range and loitering

Range = 3000 * 6076; %we're planning to fly around 3000 Nautical Miles

L_time = 45*60; %45 min loiter time, converted to seconds

endurance = Range/V_cruise;

fprintf('<strong>Range & Loiter Parameters :)</strong>\n')

fprintf('Range: <strong>%5.2f</strong> ft\n',Range)

fprintf('Loiter time: <strong>%5.2f</strong> seconds\n',L_time)

fprintf('Endurance: <strong>%5.2f</strong> seconds\n',endurance)

fprintf('--------------------------------------------------------------\n')

%Engine Properties - High-bypass turbofan

C_cruise = 0.5/3600; %1/s %SFC ----> Table 3.3 Aircraft Design - Raymer

C_loiter = 0.4/3600; %1/s %SFC ----> Table 3.3 Aircraft Design - Raymer

fprintf('<strong>Engine Properties - High-bypass turbofan :)</strong>\n')

fprintf('C_loiter: <strong>%e</strong>\n',C_loiter)

fprintf('C_cruise: <strong>%e</strong>\n',C_cruise)

fprintf('--------------------------------------------------------------\n')

%Solving for characteristics

SAM_area = 894+894+59+66+165+373+373+461+297+297+313+489+100; %updated wetted area from CAD

s_wet_body = SAM_area;
```

```
s_ref_sam = 856.8075;
swet_sref_updated = s_wet_body/s_ref_sam;
L_D_max = K_ld*sqrt(AR/swet_sref_updated); %loiter l/d ratio
L_D_cruise = 0.866*L_D_max; %cruise l/d ratio for jet
fprintf('<strong> L/D Calculation :)</strong>\n')
fprintf('L/D (MAX): <strong>%5.2f</strong>\n',L_D_max)
fprintf('L/D (CRUISE): <strong>%5.2f</strong>\n',L_D_cruise)
fprintf('----------------------------------------------------------------\n')
%5-stage fuel characteristics
w_10 = 0.97; %Take off
w_21 = 0.985; %Climb
w_32 = exp(-(Range*C_cruise)/(V_cruise*L_D_cruise)); %cruise
w_43 = exp(-(L_time*C_loiter)/(L_D_max)); %Loiter
w_54 = 0.995; %landing (Raymer)
w_50 = w_10*w_21*w_32*w_43*w_54; %total weight decrease ratio
w_f0 = 1.06*(1-w_50); %fuel to weight ratio -----> 6% residual weight allowance
fprintf('<strong>5-Stage Fuel & Weight Fraction Calculation :)</strong>\n')
fprintf('weight fraction during takeoff: <strong>%5.4f</strong>\n',w_10)
fprintf('weight fraction during climb: <strong>%5.4f</strong>\n',w_21)
fprintf('weight fraction during cruise: <strong>%5.4f</strong>\n',w_32)
fprintf('weight fraction during loiter: <strong>%5.4f</strong>\n',w_43)
fprintf('weight fraction during landing: <strong>%5.4f</strong>\n',w_54)
fprintf('weight fraction (total weight decrease ratio): <strong>%5.4f</strong>\n',w_50)
fprintf('weight fraction (fuel with 6-percent residual weight allowance): <strong>%5.4f</strong>\n',w_f0)
fprintf('----------------------------------------------------------------\n')
%-----------------------------------------------------------------------------
%% determining takeoff weight (Sensitivity analysis)
```

```
a = 0.32; %table 6.1
b = 0.66; %table 6.1
C1 = -0.13;
C2 = 0.3;
C3 = 0.06;
C4 = -0.05;
C5 = 0.05;
K_vs = 1.0; %constant sweep
guess_to_weight = 12500; %lb %initial weight guess from the Twin Otter
w_drop = 0;
rhs = 1;
t_w_ratio = 1.18; %can be 1.18 depending on the iteration we use
w_os = 45; %wing loading
M_max = M_cruise; %recalling mach number
tolerance = abs((1 - (guess_to_weight/rhs))*100); %tolerance condition
iteration = 0;
%while loop is going to iterate a solution for the empty weight of the
%aircraft
while tolerance > 0.01
   guess_to_weight = guess_to_weight + 0.1; %increasing weight by 5lb
    w_e0 = (a + ((b*guess_to_weight^C1) * (AR^C2) * (t_w_ratio^C3) * (w_os^C4) *
(M_max^C5)))*K_vs; %estimating empty weight ratio based on the initial guess
         rhs = w_pl + w_crew + w_drop + (guess_to_weight*w_f0) +
w_e0*guess_to_weight; %updating the right hand of the equation
     tolerance = abs((1 - (guess_to_weight/rhs))*100); %tolerance condition
updated
   iteration = iteration + 1;
end
%weight calculations (empty and fuel)
```

```
w_fuel = rhs*w_f0;

w_e = rhs - w_fuel;

fprintf('<strong>Weight Calculation Results below :)</strong>\n')

fprintf('The Total Take-off Weight of the aircraft is estimated at: <strong>%5.2f</strong> lbs\n',rhs)

fprintf('The Fuel Weight of the aircraft is estimated at: <strong>%5.2f</strong> lbs\n',w_fuel)

fprintf('The Empty Weight of the aircraft is estimated at: <strong>%5.2f</strong> lbs\n',w_e)

fprintf('-------------------------------------------------------------\n')

%--------------------------------------------------------------------------

%% Wing Design and Parameters

WS = w_os; %lb/ft^2 %table 5.5 transport jet value (we had to reduce it to reduce the take off distance and also that 120 from the book was not reasonable as if was of a full-scale bomber)

taper_ratio = 0.8; %estimated from the CAD Design

s_ref = rhs/WS; %reference area

wing_span = sqrt(AR*s_ref); %wingspan

%root_chord = (2*s_ref)/(wing_span*(1-taper_ratio)); %optimal root chord

%tip_chord = taper_ratio*root_chord; % optimal tip chord

%mean_chord = ((2/3)*root_chord*(1+taper_ratio+taper_ratio^2))/(1+taper_ratio);

mean_chord = s_ref/wing_span;

wing_area = s_ref;

cl_cruise = (2*guess_to_weight)/(rho_imperial*V_cruise^2*wing_area);

s_wet = swet_sref_updated*s_ref; %wing wetted surface area

fprintf('<strong>Wing Area Calculation ASSUMING Rectangular Wings  Results below :)</strong>\n')

fprintf('Estimated Coefficient of Lift (cruise): <strong>%8.4f</strong>\n',cl_cruise)

fprintf('The reference area of the wings is estimated at: <strong>%8.4f</strong> ft^2\n',s_ref)
```

```
fprintf('Estimated Wetted Surface Area of the wing: <strong>%8.4f</strong> ft^2\n',s_wet)

fprintf('Estimated Wetted Surface Area of the body: <strong>%8.4f</strong> ft^2\n',s_wet_body)

fprintf('Estimated wing span of the aircraft is: <strong>%8.4f</strong> ft\n',wing_span)

fprintf('Estimated mean aerodynamic chord of the wings is: <strong>%8.4f</strong> ft\n',mean_chord)

fprintf('Estimated Taper Ratio (Not Applied) of the wings is: <strong>%8.4f</strong>\n',taper_ratio)

fprintf('--------------------------------------------------------------\n')

%----------------------------------------------------------------------------

%% Take-off distance calculator

%%SG - initial take off roll, before rotation

alt_landing = 0;

[alt_T_land, alt_a_land, P_land, alt_rho_land] = atmosisa(alt_landing);

rho_imperial_landing = alt_rho_land/515.4;

TW = 1.187; %minimizing takeoff distance

CL_to = 1.5;

CL_max = 2.5; %jet flaps/slats/flaps/

CD0 = 0.03;

e = 0.8;

K = 1/(pi*e*AR); %K-factor from Raymer takeoff distance eqns

vstall = sqrt(2*WS/(rho_imperial_landing*CL_max)); %stall speed calculation

pseudo_speed = sqrt(2*WS/(rho_imperial_landing*CL_to));

Vto = 1.2*pseudo_speed; %take off speed calculation

%from raymer pg 673

mu = 0.05; %rolling resistance

KT = TW - mu;

KA = rho_imperial_landing/((2*WS))*(mu*(CL_to)-(CD0)-(K*CL_to));
```

```matlab
SG = (1/(2*32.2*KA))*log((KT+KA*Vto^2)/(KT));
%---------------------------------------------------------------------------
TakeOffDistance = (Vto^2)/(2*32.2*(23000/rhs - mu - 0.01));%SG*1.3... 0.01 is the D/W value
fprintf('<strong>Takeoff distance calculation :)</strong>\n')
fprintf('Estimated stall speed: <strong>%8.4f</strong> ft/s\n',vstall)
fprintf('Estimated takeoff speed: <strong>%8.4f</strong> ft/s\n',Vto)
fprintf('Estimated takeoff distance: <strong>%8.4f</strong> ft\n',TakeOffDistance)
fprintf('--------------------------------------------------------------\n')
%---------------------------------------------------------------------------
%% ------------------------------------------------------------------------
% Exam 2 starts here
%---------------------------------------------------------------------------
%% Thrust to Weight Ratio
arbi_time = endurance/2; %arbitrary half time where the plane is guaranteed be in cruise
t_w_cruise = 1/(L_D_cruise); %thrust to weight ratio during cruise
cruise_wt = w_10*guess_to_weight; %weight of aircraft during cruise
t_cruise = t_w_cruise*cruise_wt; %estimated thrust required during cruise
t_w_takeoff = t_w_cruise*w_10*(23000/t_cruise); %math checks out
fprintf('<strong>Thrust-to-weight Ratio (Takeoff) Calculation :)</strong>\n')
fprintf('Thrust to Weight Ratio (takeoff): <strong>%8.4f</strong>\n',t_w_takeoff)
fprintf('--------------------------------------------------------------\n')
fprintf('<strong>Script Ends Here For Now </strong>\n')
%---------------------------------------------------------------------------
%% Script Ends here as of now.
```

APPENDIX D - MATLAB script for Cost Analysis

```
%% Cost Analysis based on Modified DAPCA IV Cost Model
% The calculator assumes a maximum velocity of 0.8 Mach or 1125.3 ft/s with
% inlet temperature of 27 degree celcius, engine produces 13000 lb thrust
% each.
V_max = 563; % Maximum velocity of aircraft ft/s
FTA = 3; % Number of flight tests per unit
Q = 500; % Number of units made in 5 years
Neng = 2*500; % Number of engines per 500 units
t_max = 13000; % Engine maximum thrust (lb)
M_max = 0.5; % Engine maximum mach number
T_in = 2200; % Inlet temperature of engine (R)
Cav = 400000*500 % Total Avionics cost $
Re = 158; % Hourly rate of engineering $
Rt = 162; % Hourly rate of tooling $
Rm = 150; % Hourly rate of manufacturing $
Rq = 135; % Hourly rate of QC $
He = 4.86*w_e^(0.777)*V_max^(0.894)*Q^(0.163); %Engineering hours (fps)
Ht = 5.99*w_e^(0.777)*V_max^(0.696)*Q^(0.263); %Tooling hours (fps)
Hm = 7.37*w_e^(0.82)*V_max^(0.484)*Q^(0.461); %Mfg hours (fps)
Hq = 0.133*Hm; % QC hours for non cargo plane (fps)
CDevelopment = 91.3*w_e^(0.630)*V_max^(1.3); % Development support cost (fps)
Cflight = 2498*w_e^(0.325)*V_max^(0.822)*FTA^(1.21); % Flight test cost (fps)
Cmaterial = 22.1*w_e^(0.921)*V_max^(0.621)*Q^(0.799); % Manufacturing material
cost (fps)
Cengine = 3112*(0.043*t_max+243.25*M_max+0.969*T_in-2228); % Engine production
cost (fps)
RDTE_fly =
He*Re+Ht*Rt+Hm*Rm+Hq*Rq+CDevelopment+Cflight+Cmaterial+Cengine*Neng+Cav; %
Total cost for 500 units $
```